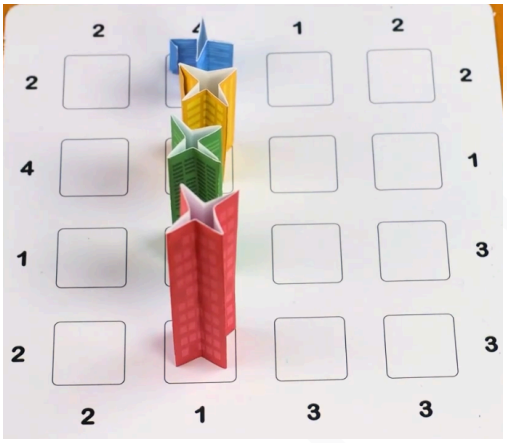
Building Puzzle

Grade 4 : Hide and Seek

Introduction

The Building puzzle is a logic-based grid puzzle where the player imagines building skyscrapers of varying heights in a city grid. The numbers outside the grid tell how many buildings are visible from that direction. Taller buildings block the view of shorter ones behind them. One can also use any objects (like blocks or bottle caps) of different heights in place of buildings to play physically.



Activity	Time	Description
Launch	5 min	The teacher introduces the puzzle with a simple example (4×4 grid) The teacher demonstrates the rules by placing some of the building blocks on the grid, and gives students a chance to try out. Template: Building: https://tinyurl.com/buildingpuzzletemplate Puzzle Grid: https://tinyurl.com/buildingpuzzletemplate2
Making and Exploration	25 min	Students now make the buildings by cutting and folding from the template and try out the challenges in the puzzle grid. Students work alongside the worksheet while the teacher guides the discussion. Student Worksheet: https://tinyurl.com/buildingpuzzleworksheet
Discussions & Conclusion	5 min	Discussing the worksheets.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

		Discuss: Strategies used, common mistakes, and multiple possible solutions Transition to: Number patterns and Magic square connection

CT Components

Algorithmic Thinking

Follow a process: Scan from the given side, track the tallest building seen so far, and count visible buildings.

Decomposition

Breaking the puzzle into smaller constraints: grid size, allowed value and given clues

Pattern Recognition

Recognising that this puzzle structure is similar to Sudoku

Generalisation

Extending to n x n grids, Generalisation pattern: $1 + 2 + \cdots + n = n (n + 1) / 2$
Compare two systems and identify similarities.

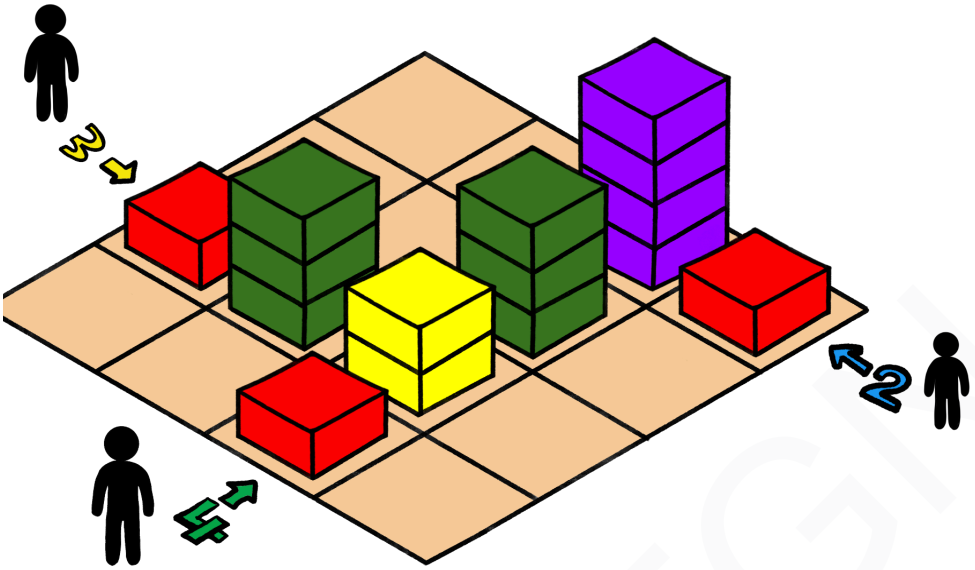
Abstraction

Physical objects represent numerical values and how the arrangement affects visibility.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Activity: Building Puzzle

Help the city planner to figure out where to build skyscrapers using clues.

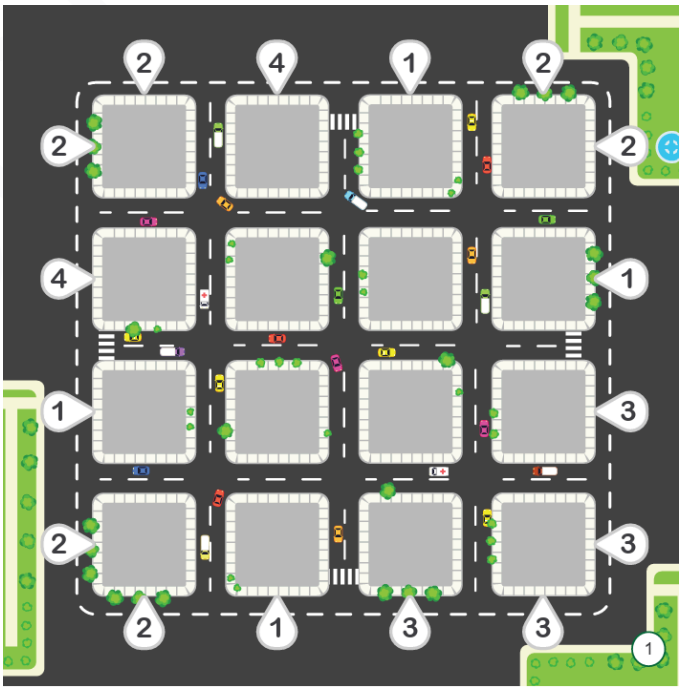


Place a skyscraper in each square. Each row and each column must contain buildings of different heights; i.e., no two buildings in a row or column have the same number of floors.

The number outside the grid (the clues) specifies how many buildings you can see if you stand on that number.

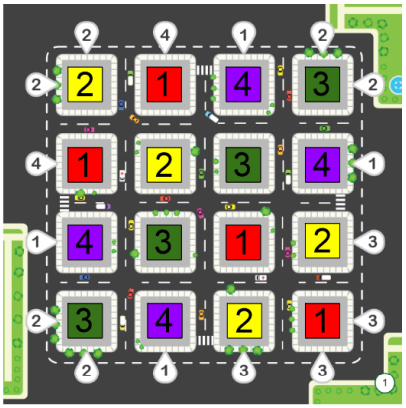
Taller skyscrapers block the view of shorter skyscrapers located behind them.

Place the skyscrapers in the grid below so that they follow all the rules of the puzzle.



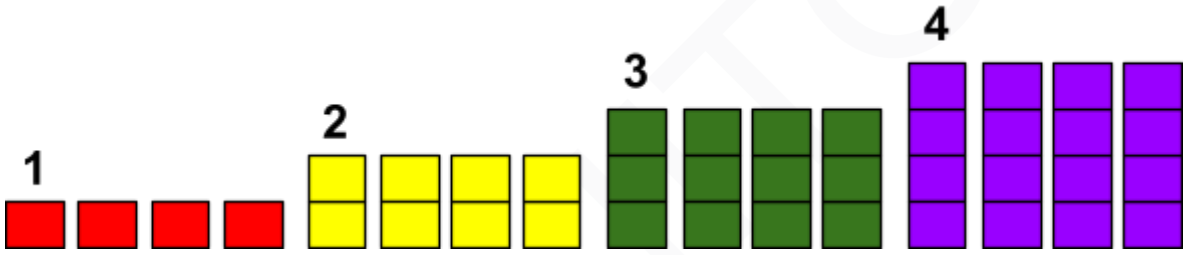
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Solution:



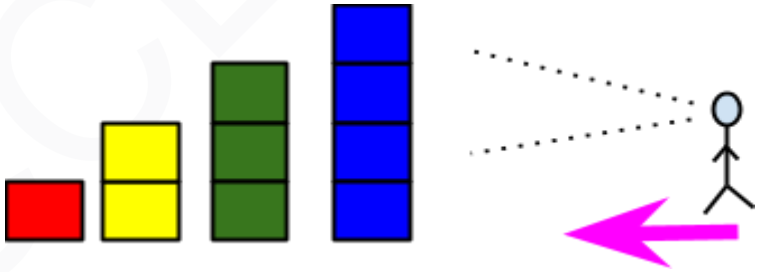
Understanding the Rules

Rule 1. Buildings of heights of 1, 2, 3, or 4 units. 4 of each different height. A total of 16 buildings are required.



If buildings are arranged in order 1, 2, 3, 4 (increasing height), all buildings will be visible from the left, since each is taller than the previous one.

Q. If buildings are arranged like 1, 2, 3, 4 (increasing height), how many buildings can you see from the right?



a) One
b) Two
c) Three
d) Four
Answer a) One
Only the tallest building blocks others. Buildings 1, 2, and 3 will be hidden behind building 4.
CT Competency: Pattern Recognition, Abstraction

Q. How many buildings are visible in this row from the left: 2, 4, 1, 3?

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

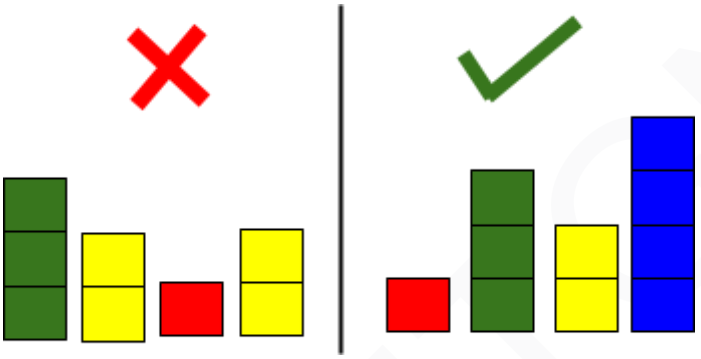
a) Four
b) Three
c) Two
d) One

Answer c) 2

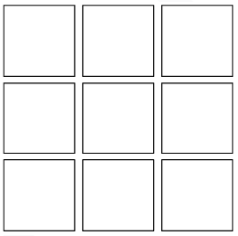
Building 2 is visible because it is the first building. Building 4 is taller than Building 2 and visible. Buildings 1 and 3 will be hidden behind Building 4, leaving only three buildings visible.

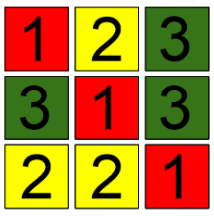
CT Competency: Algorithmic Thinking

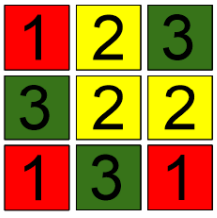
Rule 2. Buildings in each row and column must all be different heights.

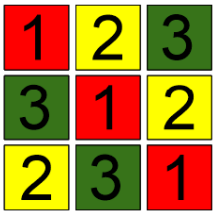


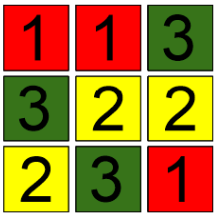
Q. What would be the correct placement for the given 3 x 3 grid if 9 buildings are placed such that every row and column follows Rule 2, for building heights 1, 2 and 3?

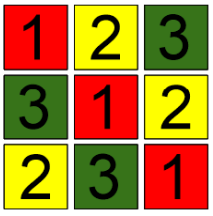


a) 

b) 

c) 

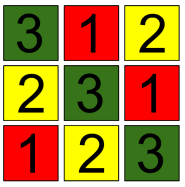
d) 

Answer c) 

The buildings in each row and column must have different heights.

CT Competency: Pattern Recognition

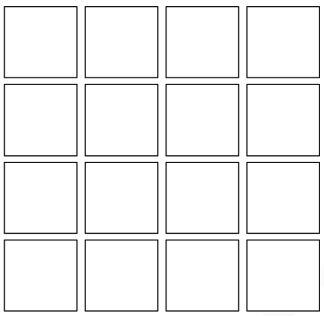
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.



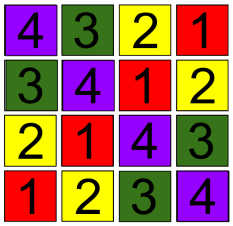
Give students a chance to observe that this will not be the unique solution.

Also, draw a similarity between this building puzzle and the Sudoku game.

Place all 16 skyscrapers in the given 4x4 grid such that every row and column follows Rule 2.



One of the solutions is shown below,



In the correct placement of both challenges, observe that the sum of all numbers in each row and in each column is equal.

In a 3x3 grid, sum is 1 + 2 + 3 = 6
In a 4x4 grid, sum is 1 + 2 + 3 + 4 = 10

Q. What should be the sum of row or column if building height is 1,2,3,4 and 5 for a 5 x 5 grid?

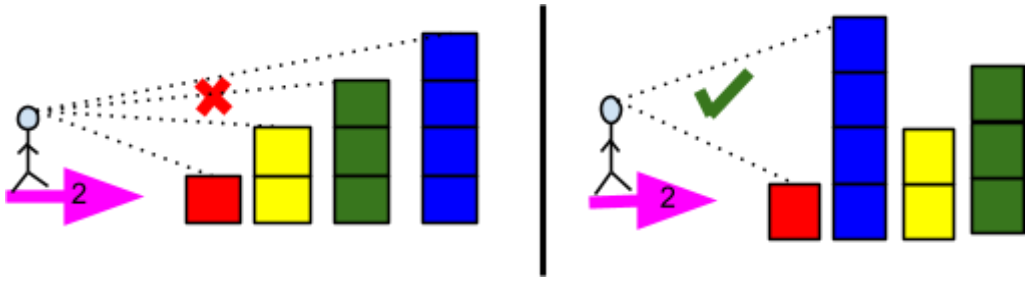
a) 6
b) 10
c) 12
d) 15

Answer d) 15
The sum of all numbers in each row in a 5x5 grid will be 1 + 2 + 3 + 4 + 5 = 15
CT Competency: Generalisation

We can generalise this sum to any grid of size n x n : 1 + 2 + ··· + n = n (n + 1) / 2

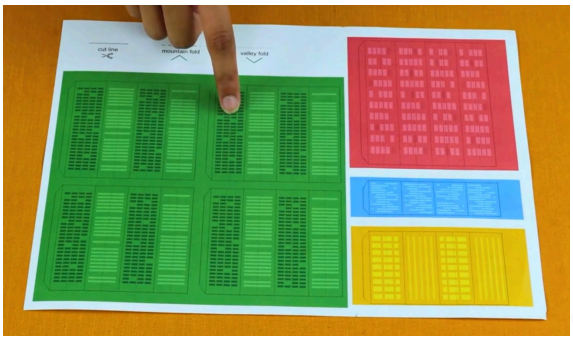
Rule 3. The number in a row tells you how many skyscrapers you should see if you were standing on that arrow.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.



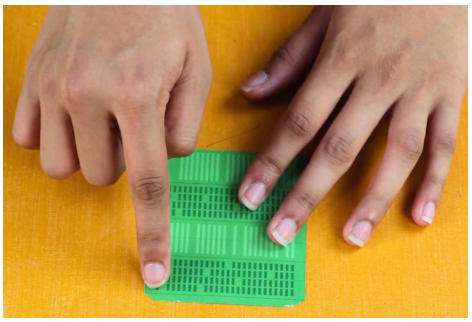
Making:

1. Take the building template.

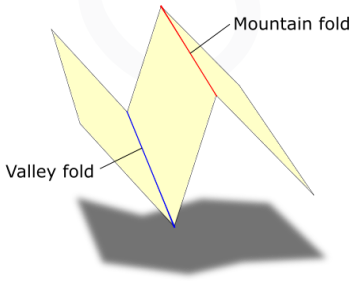


2. Cut one strip.





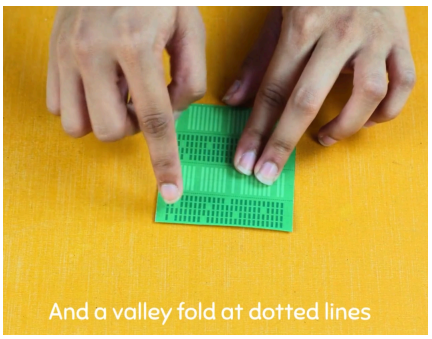
3. Understand mountain fold (like a hill) and valley fold (like a dip).



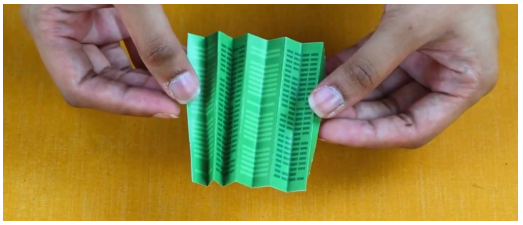
4. Make a mountain fold at the dashed lines, and a valley fold at the dotted lines.

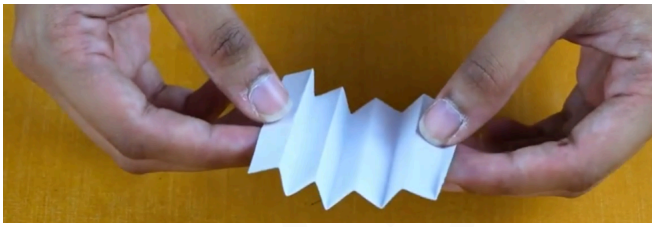
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.



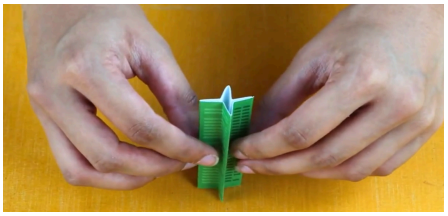


5. Continue making alternate folds like a fan.

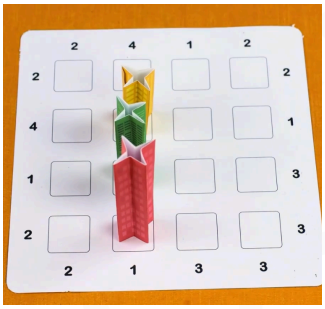




6. And your building is ready.



7. Now place the buildings in the grid. Follow all rules carefully. Try different combinations until it fits.



8. You can also use household objects to replicate the buildings.



Rules Summary

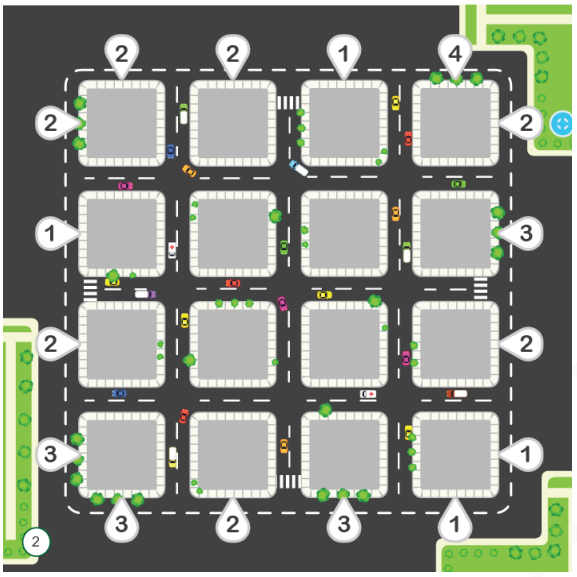
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

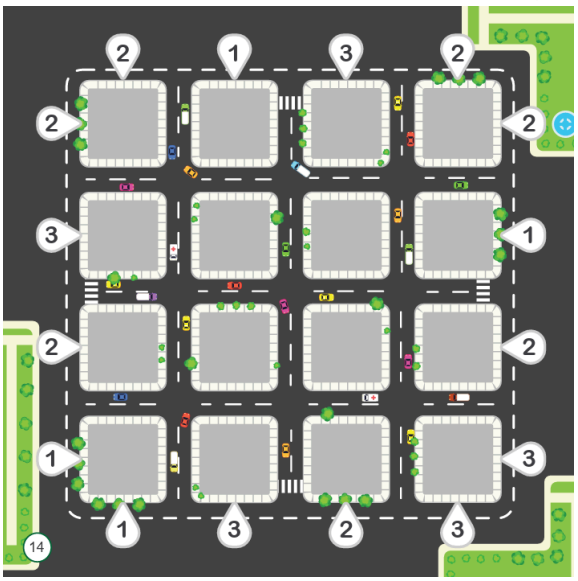
1. The heights of the buildings range from 1 to n in an n x n grid. Taller buildings block shorter ones.
2. Each row and column must have all different heights.
3. Numbers outside of the grid are a clue to show visible buildings

Follow all the rules and attempt the given challenges.

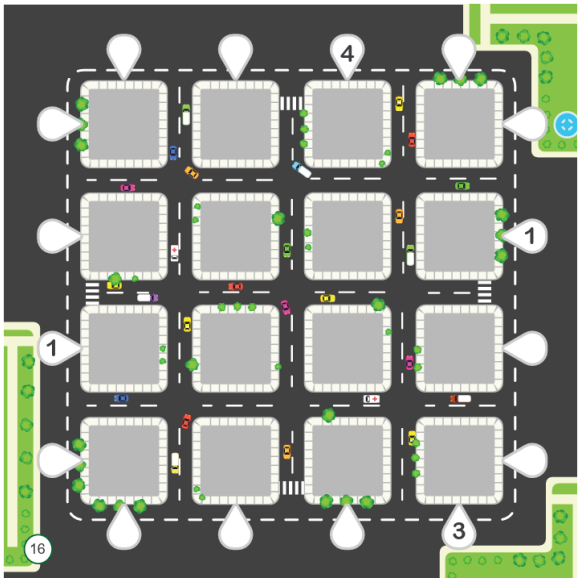
Teachers can provide more challenges within the Grid puzzle templates. The teacher can give students more time to solve the additional grids.

Challenges:

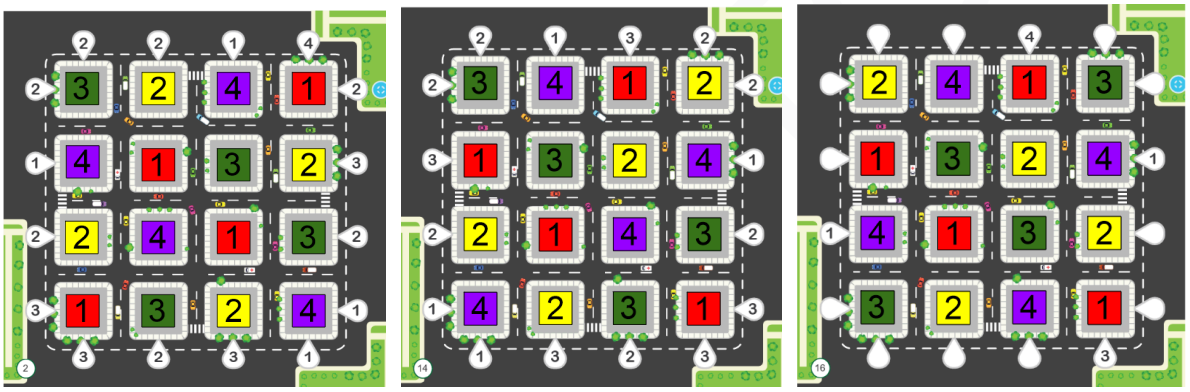




© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.



Solutions:



Additional Challenges

Q. Can you write the numbers 1 to 9 in a 3 x 3 grid (using each number once) such that the sum of every row, column, and diagonal is 15?

One of the solutions is shown below,

4	9	2
3	5	7
8	1	6

This is considered a Magic square. It is an arrangement of numbers in a square grid such that the sum of numbers in each row, column and main diagonal is equal.

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

The Building puzzle is considered a semi-magic square, where the sum of the numbers in each row and each column is the same, but the sums of the main diagonals do not necessarily equal.

Complete the given magic square. Don't use the same number twice in the square, and the numbers must add up to the same number in each row, column and diagonal line.

8		9
	6	
3		4

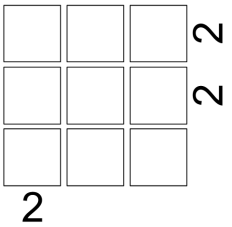
6		11
7		12

Solutions:

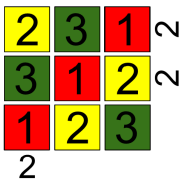
8	1	9
7	6	5
3	11	4

6	10	11
14	9	4
7	8	12

Arrange the 9 buildings of heights 1, 2 and 3 for the given 3 x 3 using the given clue.



Solution



© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.